days, providing granular intelligence that shapes strategic decisions. However, this autonomy also introduces new vulnerabilities, including dependence on satellite GPS links and exposure to electronic warfare attacks. Securing autonomy in contested communication environments remains a critical research direction for defense organizations worldwide.

Table 8.2 highlights the dual nature of UAV deployment in modern military operations, demonstrating how their tactical advantages must be carefully balanced against operational vulnerabilities. Drones offer significant benefits such as reduced personnel risk, lower operational costs, high-precision engagement, and continuous real-time surveillance, all of which enhance strategic flexibility and responsiveness on the battlefield. However, these advantages are countered by emerging threats, including cybersecurity risks, electronic warfare interference, target identification errors, and evolving counter-drone defense systems.

Table 8.2 Tactical advantages and operational risks of military UAV deployment.

Advantages of UAV use	**Operational impact**	**Associated risks/ vulnerabilities**	**Mitigation and control measures**
Reduced risk to human pilots	Allows execution of missions in high-threat environments without endangering personnel	Reliance on remote communication links, which may be disrupted	Encrypted communication channels and backup control systems
Lower operational and maintenance costs	Enables continuous surveillance and scalability of	Exposure to electronic warfare, including jamming and signal interception	Hardened GPS systems, anti-jamming INS navigation, and